

aether_pad — case dimensions reference

Single-source dimensional data for designing the Fusion 360 case. **Verify critical dimensions with calipers before committing extrusions** — some values below are from datasheets/typicals and may differ slightly from the actual hardware on your bench.

1. Arduino Giga R1 WiFi (base board)

Property	Value
PCB outline	101.52 × 53.3 mm
PCB thickness	1.6 mm
Form factor	Arduino Mega-compatible footprint
Mounting holes	4 × M3 (Mega-compatible) — NOT used in this build (Giga hangs from Shield via stacked headers)
USB-C connector	top edge (long side), centred near corner
Power jack	top edge, barrel 2.1mm centre-positive (5.5mm OD), accepts 7-24 V
Headers	4 stacks (digital, analog, power, SDA/SCL) — 8.5 mm tall
Stack height to shield top of headers	~14 mm above PCB

Verify: mounting hole XY positions relative to PCB corner (0,0).

2. Giga Display Shield (mates on top of Giga)

Property	Value
Shield PCB outline	106 × 80 mm (measured 2026-05-03)
Display module (LCD + cover glass, visible extent)	100 × 58 mm (measured 2026-05-03)
Active pixel area	84 × 50.5 mm (800 × 480 px, 5:3 aspect, inside the display module)

Active display offset from PCB left edge	6.3 mm (measured)
Active display offset from PCB right edge	15.7 mm (computed: $106 - 6.3 - 84$)
Active display offset from PCB top edge	16 mm (measured 2026-05-03)
Active display offset from PCB bottom edge	13 mm (measured 2026-05-03)
Active display height (computed)	51 mm ($80 - 16 - 13$, matches 50.5 mm spec)
Vertical bias of active within PCB	~1.5 mm toward the bottom
Mounting holes	4 corners, M3, 5 × 5 mm inset from each PCB edge (measured)
Mounting hole pattern (centre-to-centre)	96 × 70 mm — these are the only case mounting bosses needed
Front-panel window cutout	100 × 58 mm to expose the full display module
USB-C connector (on shield)	edge OPPOSITE the 6.3 mm-margin side (right edge in default orientation)
Total stack height (Giga + headers + shield)	~22 mm from Giga PCB bottom to shield top surface

IMPORTANT for assembly: mount the shield so its **6.3 mm-margin edge faces the encoder**. The opposite edge has 15.7 mm of dead PCB beyond the active area, and orienting that side toward the encoder would cost ~9.4 mm of clearance. The flip-mode rotation preserves this — when the top rotates 180°, the 6.3 mm edge still faces the (now opposite-side) knob.

3. Autonics E50S8G5-360B-G24N optical encoder

Property	Value
Body diameter	50 mm
Body length (behind panel)	38.6 mm (measured 2026-05-03)
Total length incl. shaft	~52 mm
Shaft diameter	8 mm

Shaft length	~13 mm beyond body
Mounting	3 × M3 screws on a 40 mm PCD (20 mm radius), on the front face — measured
Panel mount hole	32 mm centre clearance (to clear shaft + boss)
Wiring	5-wire pigtail: A, B, Z, +V (12-24V), 0V
Operating voltage	12-24 V DC (uses 13.8 V from shack rail)
PPR	360

For the recessed-knob design:

- Panel cutout: 32 mm hole for shaft + boss, 3 × M3 tapped or clearance holes on **40 mm PCD**
- Behind-panel clearance needed: **38.6 mm minimum depth** (encoder body length), 50+ mm diameter
- Knob hollow cavity: 52 mm ID, 38.6+ mm deep so it clears the encoder body
- Knob outer dimensions: ~55-60 mm OD typical, 10-15 mm tall above panel
- Knob coupling: 8 mm bore, M3 grub screw onto shaft flat

4. CW paddle jack — 3.5 mm TRS, panel mount

Property	Value
Panel hole	6 mm
Body length behind panel	~15-20 mm (varies by part)
Mounting	nut + lock washer on threaded barrel
Wiring	tip = dit (D4), ring = dah (D5), sleeve = GND

5. Power entry — Anderson PowerPole, daisy-chainable

12 VDC in via Anderson PowerPoles. **Two pairs** (red+black, red+black) on the rear panel — one pair is the feed in, the second pair allows daisy-chaining the next shack module on the same supply. Dimensions confirmed against the parts on hand.

Property	Value
Housing size (single PP15/30/45)	7.87 × 11.94 × 16.26 mm

Red+black pair (clipped together)	15.75 × 11.94 × 16.26 mm
Cutout per pair	16.3 × 18.4 mm (snug fit on housing, ~2.7 mm clearance for retainer wall)
Retainer sleeve	24.6 mm long, 3 mm wall thickness around the opening — 3D-printed, extends into the case interior
Total clearance behind panel	~25 mm for housing + wire bend
Wiring	Wire both pairs in parallel internally (same +/- nodes) so daisy-chain just passes through

Encoder needs 12-24 V (uses 13.8 V from shack rail); Giga accepts 7-24 V. Single 12 V supply works for both.

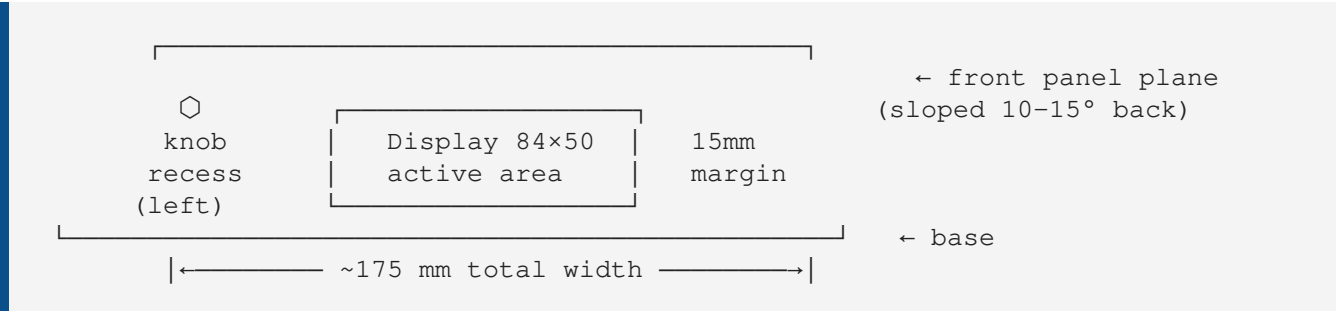
Cutout layout — TBD, two options:

- **(a)** Two separate 16.3 × 18.4 cutouts with 5-10 mm gap between → flexible, two independent retainers
- **(b)** One combined 16.3 × ~37.3 mm cutout → single retainer block, more compact

Place on rear panel, central horizontally, ~10 mm up from the case base for cable clearance.

6. Combined stack — case envelope (symmetric flip-mode, locked)

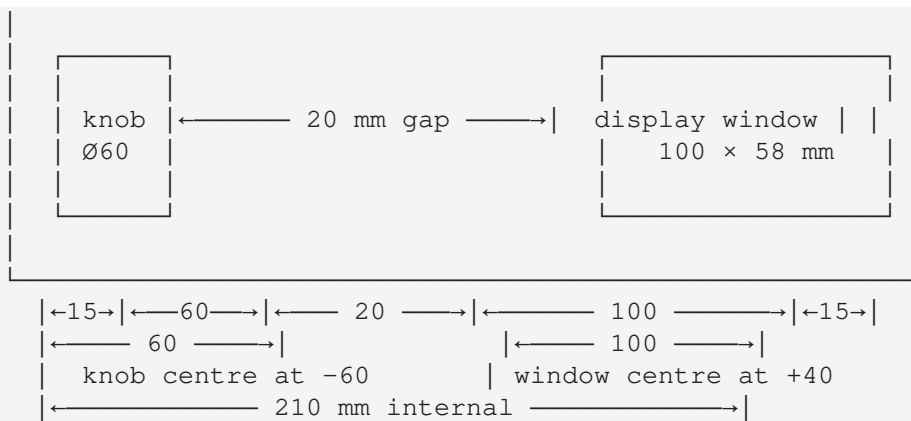
Sloped desktop unit, screen on the right of the front panel, encoder + recessed knob on the left, no side cutouts (no USB-C side access — open the case for reflashing using a USB-C extension cable).



Symmetric design — knob and screen window mirror about the case centreline. Hardware (Giga + Shield + encoder) lives in the **removable top half**, so rotating the top 180° physically moves them to the opposite side; the base stays put with rear connectors fixed in position.

Front panel layout (default orientation, encoder on left):



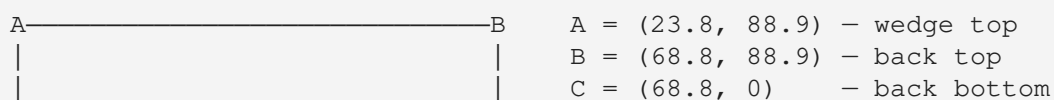


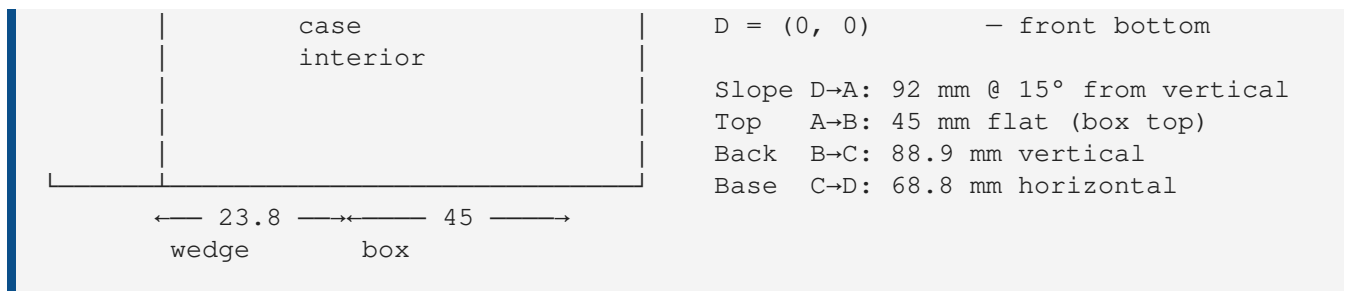
Property	Value
Margin: knob outer edge to left wall	15 mm
Margin: window outer edge to right wall	15 mm
Gap between knob and window	20 mm (gives 26.7 mm encoder-PCB clearance)
Knob centre offset from case centreline	60 mm (left in default)
Window centre offset from case centreline	40 mm (right in default)
Top + bottom margins to window	17 mm each (gives ~5 mm PCB clearance)

Internal envelope (locked):

Dimension	Value	Driver
Width	210 mm	symmetric about centreline, 60+30+15 each side, 20 gap, screen window 100
Height (perp to slope)	92 mm	window 58 + 17 top + 17 bottom margins (PCB ~5 mm clearance)
Slope angle from vertical	15°	nearly upright, compact, comfortable bench viewing
Total interior depth (front to back at base)	68.8 mm	wedge 23.8 + box 45 (box bumped to clear deeper-than-spec encoder body)
Total interior height (vertical)	88.9 mm	$= 92 \times \cos(15^\circ)$

Side profile (right trapezoid, looking from the right):





External envelope with 3 mm walls: **216 W × 75 D × 95 H mm**.

Component placement inside (verified, no 3D conflicts):

Component	Front-face / mount position	Extent into case
Display Shield + Giga stack	parallel to slope, mounted to fascia	22 mm perpendicular to slope
Encoder body	slope midpoint (X=11.9, Y=44.4), Z=-60 from centreline	38.6 mm along slope normal → back face at (X=49.2, Y=34.4); back-top corner at (X=55.7, Y=58.6)
PowerPole sleeves (×2)	back wall, centred Z, centred Y	24.6 mm into case (X from 44.2 to 68.8)
Paddle jack	back wall, centred Z	~15-20 mm sleeve
Flip switch	back wall, near corner	tiny

The encoder body and PowerPole sleeves overlap in side view (both at similar X), but they're separated in the case-width (Z) direction by ~26 mm — encoder at Z=-60, PowerPoles centred at Z=0. No 3D conflict.

Encoder back-top to back wall: 13.1 mm clearance — enough for the 5-wire pigtail strain relief and a 90° cable bend toward the base's central connector.

Box top (the 45 mm flat region between wedge apex and back wall) is useful internal volume for routing the cable between the rotating fascia and the base's rear connectors.

PCB position note: because the active display is biased ~1.5 mm toward the bottom of the PCB, when you align the display module with the case-centred window, the PCB itself sits ~1.5 mm UPWARD of the case vertical centre. Standoff heights need to account for this if you're putting the PCB on a centred mount.

After 180° top rotation: knob ends up on the right at +60, window on the left at -40, all margins preserved. Identical clearances in both orientations.

Front panel artwork (default orientation, encoder on left)

Front panel size: **210 × 92 mm** (in the plane of the slope). Origin at lower-left, X right, Y up.

Visible features (cuts through the front face):

Feature	Position	Size / type
Screen window cutout	centre (145, 46)	100 × 58 mm rectangle, 2-3 mm corner radius
Knob shaft clearance	centre (45, 46)	Ø32 mm round cutout
Encoder mount #1 (12 o'clock)	(45.0, 66.0)	Ø3.5 mm M3 clearance
Encoder mount #2 (4 o'clock)	(62.3, 36.0)	Ø3.5 mm M3 clearance
Encoder mount #3 (8 o'clock)	(27.7, 36.0)	Ø3.5 mm M3 clearance

The 3 encoder mounting holes are 120° apart on a **Ø40 mm PCD** (20 mm radius), measured 2026-05-03. The orientation shown puts one hole at 12 o'clock — rotate the trio if the encoder pigtail needs clearance elsewhere.

Internal-only features (back side of fascia, no through-holes):

Feature	Position	Type
Shield mounting standoff	(101.7, 82.5)	M3 self-tap or threaded insert
Shield mounting standoff	(197.7, 82.5)	(4 corners of shield PCB)
Shield mounting standoff	(101.7, 12.5)	
Shield mounting standoff	(197.7, 12.5)	

Standoff heights set the gap between fascia inner surface and PCB top — typically 5 mm to clear the LCD module thickness.

No screws on the front face — the fascia attaches via an inner skirt + side screws (see §7).

7. Two-half case construction (the rotating-top design)

The case is split horizontally:

- **BASE** = standard, fixed orientation. Rear panel with connectors. Sits on bench. Side walls have screw-engagement holes for the fascia skirt.
- **TOP (fascia)** = removable, contains all active hardware (Giga + Display Shield + encoder + their wiring). Has an **inner skirt** on the underside that drops into the base interior; held captive by **side screws** through the base side walls into the skirt. No screws visible on the front face.

To flip for friend-mode: power off, undo side screws, lift top off, unplug internal cable, rotate top 180°,

reseat, replug, rescrew, set the rear flip switch, power up. ~60 seconds.

Mechanical requirements (preserves flip-mode rotational symmetry):

- Inner skirt outline: **rectangular, centred on the case**, slightly smaller than the base interior (~2 mm under-size all round for a snug locate fit). Skirt depth into base ~10-15 mm.
- Side screws on **mirrored positions left and right** at the same height — typically 2 per side at top and bottom of the engagement region.
- Internal cable connector (5 wires: +12V, GND, paddle dit, paddle dah, flip switch) located on the **case centreline** so both orientations reach it.
- Rear connectors stay on the base, position-independent of top rotation.
- Top must be wide and deep enough internally to hold the hardware in either orientation without collision (drives the 210 × 92 × 53.8 mm interior dimensions in §6).

8. Firmware changes needed for flip-mode

Add to `aether_pad.ino`:

```
const int FLIP_PIN = 6;           // rear switch → D6, INPUT_PULLUP
bool flipped = false;             // set in setup() from FLIP_PIN

// In setup(), after pinMode/INPUT_PULLUP on FLIP_PIN:
flipped = (digitalRead(FLIP_PIN) == LOW);
tft.setRotation(flipped ? 3 : 1);
touchMode = flipped ? 1 : 0;      // mode 1 = 180° from default

// In encISR, conditionally invert direction:
encRaw += (flipped ? -1 : +1) * QUAD[(encLast << 2) | now];
```

That's it — ~5 lines plus a `pinMode` in `setup`.

9. USB-C access — decided: no external cutout

No external USB-C access on the case. For reflashing, open the case (or use a USB-C extension cable threaded through one of the rear cutouts temporarily). This keeps the side panels clean, simplifies symmetry for flip-mode, and avoids needing a panel-mount USB-C extension.

10. Rear panel — connector layout summary

All external connectors live on the rear panel. Suggested layout (left to right when viewed from the back):

[PP×4 cutout] 32×17 mm	[3.5mm jack] Ø6 mm	[flip switch] if symmetric
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[vent slots]

Connector	Position	Cutout
PowerPole pair × 2	rear panel, two separate cutouts with ~5-10 mm gap between	16.3 × 18.4 mm each (24.6 mm sleeves into case)
3.5 mm CW paddle TRS	rear panel, central	ø6 mm circular
Flip-mode switch	rear panel, near corner	small SPST toggle, ~6 mm hole
Ventilation slots	rear panel, lower portion	array of slots, ~10% open area

10. Ventilation

The Giga R1 runs warm (STM32H747 dual-core), especially with WiFi active. Recommend:

- Slot pattern on bottom panel under the Giga
- Slot pattern on rear panel (or wherever the convection exit is on a sloped enclosure)
- 3-5 mm wall-to-PCB-bottom standoff so air can flow under

Total open area ~10% of the relevant face is plenty for passive cooling.

TODO before final case design

- [x] ~~Caliper-verify Giga mounting hole XY pattern~~ → not needed; Giga hangs from Shield via headers. Optional printed support clips between Giga and Shield to take bending load off the header pins.
- [] Caliper-verify shield active-display XY relative to mounting holes
- [] Caliper-verify shield mic position (if grille wanted on top of case)
- [x] ~~Decide power entry style~~ → Anderson PowerPole, 16.3 × 18.4 mm per pair, 24.6 mm sleeve, 3 mm wall
- [x] ~~Decide USB-C strategy~~ → no external access, open case for reflash
- [x] ~~Decide screen-edge to side-of-case margin~~ → 15 mm right side
- [x] ~~Decide encoder offset~~ → 60 mm from case centreline (default = left), vertically centred
- [x] ~~Decide PowerPole cutout layout~~ → two separate 16.3 × 18.4 cutouts, ~5-10 mm gap between
- [x] ~~Confirm knob OD~~ → 60 mm assumed (verify when knob is sourced/printed)
- [x] ~~Commit to flip-mode~~ → YES, via removable+rotatable top half (see §7)
- [x] ~~Verify shield active display offset within PCB~~ → 6.3 mm from left, 13 mm from top
- [x] ~~Verify display module size~~ → 100 × 58 mm (front-panel window)

- [x] ~~Lock case slope angle~~ → **15° from vertical** (interior 68.8 mm deep, 88.9 mm tall, wedge front + 45 mm box back)
- [x] ~~Caliper-verify encoder body depth and mounting PCD~~ → 38.6 mm deep (was 30 in datasheet typical), 40 mm PCD (was 30) — measured 2026-05-03
- [x] ~~Source or design the recessed knob~~ → 3D-print in 2 parts with pause-print at the steel-ball layer to embed weight (Nigel's design). Confirm final OD when designed — affects case width if >60 mm.
- [] Source PowerPole panel-mount retainer (commercial or 3D-printed)
- [] Add flip-pin handling to firmware (~5 line change, waits until physical switch is wired)
- [] Wire the flip-mode switch on rear panel (D6, INPUT_PULLUP)